

Narrowing the ANN–SNN Gap for Continuous 1D Temporal Signal Classification with Multi-Scale Temporal Encoding and Sparsity-Regularized Transform Encoding

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Abstract

Spiking neural networks (SNNs) promise energy-efficient inference, yet on static vision benchmarks, they often trail matched ANNs under short simulation horizons. Under a matched-backbone and matched-budget protocol without extra tricks, we find that this ANN–SNN accuracy gap is consistently smaller on representative continuous 1D temporal signal classification benchmarks than on image benchmarks. We attribute this to a mechanism-level mismatch: leaky integration naturally implements causal evidence accumulation over time for native temporal signals, while static images typically require amplitude-to-spike encoding, whose finite-window estimation error becomes non-negligible at short horizons. Guided by this view, we propose a plug-and-play framework that combines Multi-Scale Temporal Encoding (MTE) and Sparsity-Regularized Transform Encoding (STE). MTE replaces naive repetition with multi-scale streams and allocates scale-aligned multi-bit integer spikes to increase per-step information density, and STE replaces a controllable fraction of LIF units with a transform-encoding neuron trained using auxiliary reconstruction and sparsity regularization, with a synthesis branch used only during training. Across diverse 1D datasets and backbone families, MTE×STE consistently improves the accuracy-efficiency trade-off over standard SNN baselines and matches or occasionally surpasses ANN counterparts. The source code is available at https://github.com/Sun7-7/Signal_Code_SNN

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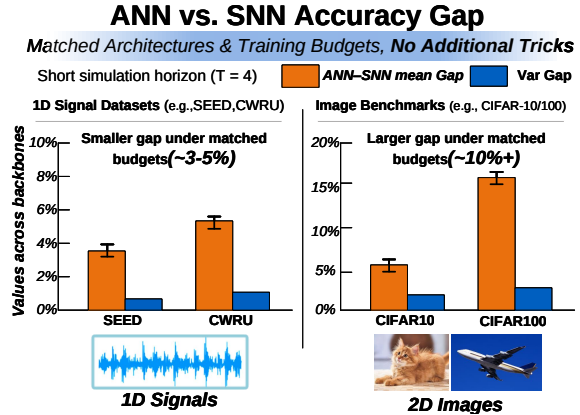


Figure 1. Under matched backbones without extra tricks, the ANN–SNN accuracy gap is substantially smaller on representative continuous 1D temporal signal than on image.

1. Introduction

Spiking neural networks (SNNs) enable event-driven computation with sparse spikes and internal state evolution, making them attractive for low-power inference on neuromorphic substrates. Often viewed as a “third-generation” neural model (Maass, 1997), SNNs can now be trained end-to-end via surrogate gradients and backpropagation through time (BPTT), and have scaled to increasingly deep architectures on vision benchmarks (Wu et al., 2019; Ding et al., 2021; Zhou et al., 2022). Nevertheless, on *static* image classification, directly trained SNNs under short simulation horizons still frequently trail matched ANNs, and narrowing the gap typically relies on longer horizons or additional training/encoding refinements (Sengupta et al., 2019; Zheng et al., 2021; Ding et al., 2021).

In this work, we revisit a practically important observation under a strict matched protocol: under matched backbones and training budgets without additional tricks, the ANN–SNN accuracy gap is smaller on representative continuous 1D temporal signal classification benchmarks (e.g., SEED and CWRU) than on static image benchmarks (Fig. 1). The phenomenon demonstrates a situation where time structure must be artificially induced from static amplitudes, whereas

our target setting is native continuous-time evidence. This matters for temporal recognition in EEG/ECG/EMG and speech, as well as industrial diagnostics, where continuous 1D temporal signals exhibit structured temporal dynamics and sparsity (Song et al., 2018; El Bouny et al., 2020; Zhang et al., 2019; 2023). Why does the gap shrink on native temporal signals? We posit a neurodynamics-driven hypothesis: leaky membrane integration implements causal memory with an interpretable time constant, aligning with temporal evidence accumulation in continuous 1D temporal signals (Gerstner & Kistler, 2002; Dayan & Abbott, 2005). In contrast, short-horizon SNN methods for static images often rely on amplitude-to-spike encoding, where finite windows can induce non-negligible information loss and uncertainty (Yan et al., 2025; Carbajal & Romero, 2024). We formalize this hypothesis as a unified proposition in Section 4.1 and derive implications for short-horizon, energy-aware spiking classification.

Guided by this analysis, we propose a plug-and-play framework coupling Multi-Scale Temporal Encoding (MTE) and Sparsity-Regularized Transform Encoding (STE) to improve the accuracy–energy trade-off under short horizons (Fig. 2). MTE replaces naive repetition with multi-scale temporal encoding and allocates scale-aligned multi-bit integer spikes to match information density across scales, using integer-valued neurons that can emit multiple spikes per time step (Feng et al., 2022; Xu et al., 2025). STE is a transform-encoding neuron plugin that replaces a controllable fraction of LIF units and is trained with auxiliary reconstruction and sparsity regularization, while the synthesis branch is used only during training (thus not increasing inference cost). Unlike prior work that primarily scales model complexity or relies on fixed encoders and membrane parameters for temporal modeling (Fang et al., 2020; 2021; Lv et al., 2024; Dominguez-Morales et al., 2018), our design targets short-horizon information efficiency and controllable firing activity via standard spike-based energy proxies, yielding a favorable accuracy-energy Pareto frontier across diverse 1D datasets and backbone families (e.g., RNN/TCN/CNN/Transformer).

- **Matched-budget gap observation.** Under matched backbones and training budgets (without additional tricks), we show the ANN-SNN gap is typically smaller on representative continuous 1D temporal signal benchmarks than on static image benchmarks.
- **Unified mechanism-level analysis.** We formulate a unified proposition linking leaky integration (time-constant memory) and finite-horizon encoding uncertainty, and derive short-horizon design principles for temporal signals.
- **MTE×STE framework.** We propose Multi-Scale Temporal Encoding (MTE) and Sparsity-Regularized

Transform Encoding (STE) to improve the accuracy-energy trade-off across datasets and backbone families.

2. Related Work

2.1. Deep SNNs and short-horizon encoding

Surrogate-gradient training with BPTT has enabled end-to-end optimization of SNNs and their scaling to deep architectures (Wu et al., 2019; Sengupta et al., 2019; Zheng et al., 2021; Ding et al., 2021; Zhou et al., 2022). Although SNNs can be competitive with sufficient timesteps and carefully tuned recipes, short-horizon inference on *static* images often remains sensitive to input-to-spike encoding fidelity. In contrast, event-based vision with native temporal structure typically favors spiking computation (Li et al., 2017; Amir et al., 2017). Multi-level/multi-bit firing has also been explored to improve representation capacity and enable deeper directly-trained SNNs (Feng et al., 2022; Xu et al., 2025). Our work targets the matched-budget short-horizon regime and couples (i) scale-aligned multi-bit precision allocation at the encoder with (ii) neuron-level firing regulation via a plug-in transform-encoding module, featuring a training-only synthesis branch and a controllable replacement ratio.

2.2. SNNs for continuous 1D temporal signal recognition

SNNs have been applied to diverse 1D temporal recognition tasks, including EEG (Song et al., 2018; Phyo et al., 2022; Bi et al., 2024), ECG (El Bouny et al., 2020; Prabhakararao & Dandapat, 2021), modulation recognition (Zhang et al., 2023), mechanical fault diagnosis (Zhang et al., 2019; Wang et al., 2022), and speech/audio classification. Existing studies typically improve backbones or training pipelines, while efficiency is commonly reported using spike activity and operation-count proxies. We complement these efforts by focusing on a matched-budget ANN-SNN gap phenomenon and proposing a short-horizon design that allocates representational precision across temporal scales and regulates firing activity within backbones, yielding improved accuracy-energy Pareto trade-offs.

3. Preliminaries

Problem formulation under a matched protocol. We study short-horizon SNN inference for continuous 1D temporal signal classification under a matched-backbone, matched-budget protocol. Given an input segment $x \in \mathbb{R}^{C \times L}$ (channels C , length L) and a label $y \in \{1, \dots, K\}$, the goal is to learn a predictor $f(\cdot)$ that outputs $\hat{y} = f(x)$. We denote a mini-batch by $X \in \mathbb{R}^{B \times C \times L}$. To isolate the effect of spiking computation and our proposed components, we compare each ANN backbone and its SNN counterpart under the same macro-architecture, data preprocessing, optimizer family, number of training epochs, early-stopping rule,

batch size, and per-run training budget, without additional tricks or extra model-selection budget.

Short-horizon spiking computation. We adopt a discrete-time SNN with a short simulation horizon T . Let $\{X^{(t)}\}_{t=1}^T$ denote the T -step input stream fed into the spiking backbone, where $X^{(t)} \in \mathbb{R}^{B \times C \times L}$. A spiking neuron maintains a membrane state u_t and emits spikes s_t . A generic leaky integrator with reset can be written as:

$$u_t = \alpha u_{t-1} + I_t - \vartheta s_{t-1} \quad (1)$$

where $\alpha \in (0, 1)$ is the discrete decay factor corresponding to an interpretable time constant, I_t is the synaptic input (e.g., a linear transform of $X^{(t)}$), and ϑ controls the reset strength. Neurodynamically, Eq. (1) can be viewed as a discretization of a leaky integrator in the subthreshold regime:

$$\tau_m \frac{du(t)}{dt} = -u(t) + I(t), \quad \alpha = e^{-\Delta t / \tau_m} \quad (2)$$

where τ_m is the membrane time constant and Δt is the simulation time step. Thresholding and reset introduce non-linearity beyond Eq. (2).

Activity and energy proxies. To quantify sparsity and efficiency, we report spike-based activity measures and synaptic operation proxies commonly used in SNN literature. Given spikes $\{s_{t,\ell}\}$ at layer ℓ and time t , we define the average spike activity (activation rate) as:

$$\text{SR} \triangleq \frac{1}{|\Omega|} \sum_{(t,\ell) \in \Omega} |s_{t,\ell}| \quad (3)$$

where Ω indexes all spiking units across time and layers. We also report synaptic operations (SOPs) that scale with spike activity and synaptic fan-in. For a spiking layer ℓ with N_ℓ neurons and fan-in d_ℓ (number of synapses per neuron), a commonly used proxy is Eq. (4) and $\text{SOPs} = \sum_\ell \text{SOPs}_\ell$. The exact accounting and its use in our activation-rate/power analysis are provided in Appendix E.

$$\text{SOPs}_\ell \triangleq \sum_{t=1}^T \sum_{i=1}^{N_\ell} |s_{t,\ell,i}| d_\ell \quad (4)$$

4. Methods

Fig. 2 shows our plug-and-play framework for short-horizon SNN classification, which couples Multi-Scale Temporal Encoding (MTE) and Sparsity-Regularized Transform Encoding (STE). MTE replaces naive repetition by generating a T -step multi-scale input stream with scale-aligned multi-bit integer spikes, while STE can replace a controllable fraction p of LIF units to regulate firing activity via sparsity-regularized transform encoding. STE employs a synthesis branch only during training for auxiliary reconstruction and sparsity regularization, incurring no extra inference cost.

4.1. Problem Formulation and Theoretical Motivation

We study short-horizon (T -step) SNN inference under the matched protocol in Section 3. Most SNN classifiers aggregate spike activity across time (e.g., rate/sum readout); hence, the information available for decision making is constrained by both (i) the neuron dynamics (Eq. (1)) and (ii) how an input is represented over a finite horizon T . A key distinction is that native temporal signals provide a genuine stream $\{I_t\}_{t=1}^T$ (e.g., samples/features over time), while static inputs lack native temporal structure and are commonly represented via spike-based encoding (e.g., rate/latency) or repetition under short horizons (Yan et al., 2025; Carbajal & Romero, 2024). Below we formalize how these two regimes interact differently with leaky integration and derive short-horizon design implications.

Linearized leaky integration. Ignoring reset for analysis (details in Appendix A), the subthreshold dynamics in Eq. (1) reduce to a causal leaky integrator:

$$u_t = \alpha u_{t-1} + I_t, \quad \alpha \in (0, 1) \quad (5)$$

The closed-form solution Eq. (6) is an exponential moving sum. It shows that leaky integration implements a stable causal memory with an effective timescale controlled by α (or τ_m in Eq. (2)) (Gerstner & Kistler, 2002; Dayan & Abbott, 2005).

$$u_t = \sum_{k=1}^t \alpha^{t-k} I_k \quad (6)$$

The signal and encoding noise. The synaptic drive can be viewed as a clean component plus two noise sources:

$$I_t = \bar{I}_t + \epsilon_t^{(\text{in})} + \epsilon_t^{(\text{enc})} \quad (7)$$

where $\epsilon_t^{(\text{in})}$ captures input/measurement noise intrinsic to the underlying signal, while $\epsilon_t^{(\text{enc})}$ captures encoding-induced noise introduced when a non-temporal stimulus (e.g., static amplitudes) is converted into a finite-horizon spike stream.

Crucially, the two input regimes differ in whether the clean drive $\bar{I}_t$ carries informative temporal structure within the same horizon. For native temporal signals, $\{\bar{I}_t\}$ itself can provide task-relevant evidence over time, such as bursts, phase/frequency cues, or local temporal patterns. For static inputs, by contrast, $\bar{I}_t$ is typically time-invariant and the temporal axis is mainly used to temporalize amplitudes, which introduces an additional finite-window encoding component $\epsilon^{(\text{enc})}$ under short horizons (Yan et al., 2025; Carbajal & Romero, 2024).

Proposition 1: Under short horizons, native 1D temporal signals are more compatible with leaky neurodynamics than static 2D inputs. Consider Eq. (5) with $u_0 = 0$ and

dence. We therefore design MTE as a front-end encoder that converts a single 1D segment into a T -step multi-scale drive stream, so that different time steps carry complementary evidence at different temporal resolutions.

Given an input mini-batch $X \in \mathbb{R}^{B \times C \times L}$, MTE outputs a T -step stream $\{X^{(t)}\}_{t=1}^T$ with $X^{(t)} \in \mathbb{R}^{B \times C \times L_t}$, which is fed into the spiking backbone as the synaptic drive I_t in Eq. (1). We define a monotonically non-increasing scale schedule $\{L_t\}_{t=1}^T$ from the true length L with a minimum resolution $L_{\min} = 32$:

$$L_t \triangleq \max(\lfloor L/2^{t-1} \rfloor, L_{\min}), t = 1, \dots, T \quad (11)$$

Let $\mathcal{D}_t : \mathbb{R}^{B \times C \times L} \rightarrow \mathbb{R}^{B \times C \times L_t}$ denote a direct downsampling operator, and let $\mathcal{U}_t : \mathbb{R}^{B \times C \times L_t} \rightarrow \mathbb{R}^{B \times C \times L}$ denote a deterministic upsampling/resampling operator. In addition to direct downsampling, MTE includes a lightweight feature-refinement branch to inject non-linear transformations Z_t between adjacent scales, where $Z_t \in \mathbb{R}^{B \times C \times L_t}$ and $Z_1 \triangleq \mathcal{D}_1(X)$.

$$Z_{t+1} \triangleq \mathcal{D}_{t+1}(X) + \mathcal{F}_t(Z_t), \quad t = 1, \dots, T-1 \quad (12)$$

The direct path $\mathcal{D}_{t+1}(X)$ preserves scale-specific signal fidelity, while the refinement path $\mathcal{F}_t(Z_t)$ injects complementary coded evidence carried from the previous step. Finally, each Z_t is upsampled back to length L and stacked along the time axis:

$$X^{(t)} \triangleq \mathcal{U}_t(Z_t), \quad \text{MTE}(X) \triangleq \{X^{(t)}\}_{t=1}^T \quad (13)$$

A distinctive feature of MTE is that it allocates representational precision across scales under a fixed short-horizon budget. Specifically, the feature-refinement branch applies an integer-valued multi-spike nonlinearity $\phi_{m_t}(\cdot)$ with a scale-dependent spike budget $m_t \in \mathbb{N}$, producing integer spike codes $S_t \in \{0, 1, \dots, m_t\}^{B \times C \times L_{t+1}}$ (Feng et al., 2022; Xu et al., 2025). These codes are then mapped to the next scale to form a refined real-valued drive; hence, the backbone always receives a real-valued drive stream $X^{(t)}$. Formally, we instantiate:

$$\mathcal{F}_t(Z_t) \triangleq \mathcal{P}_{t \rightarrow t+1}(S_t), \quad S_t \triangleq \phi_{m_t}(\mathcal{H}_t(Z_t)) \quad (14)$$

where $\mathcal{H}_t$ is a lightweight local transform at scale L_t and $\mathcal{P}_{t \rightarrow t+1}$ maps the spike codes to resolution L_{t+1} . We keep these projections lightweight by applying them per channel at reduced temporal scales, and report their MACs separately to avoid hiding dense front-end costs. In practice, we use a decreasing spike-budget schedule, allocating more precision to higher-resolution steps and less to lower-resolution steps. This scale-aligned multi-bit allocation implements Principle (i) from Section 4.1: enriching temporal evidence per step by distributing precision across scales under short horizons.

Although MTE uses integer multi-bit spike codes $S_t \in \{0, \dots, m_t\}$ inside the refinement branch, these codes can be interpreted as event counts within a time step (equivalently, multiple binary spikes emitted by a small set of virtual neurons), so synaptic computation still scales with the number of emitted events and we treat $|S_t|$ consistently in SR/SOPs as event counts. Importantly, MTE is not a feature pyramid for multi-level semantic fusion: it explicitly treats the simulation horizon as an information budget and constructs a T -step drive stream where each step corresponds to a distinct temporal resolution with an explicit spike budget m_t , targeting short-horizon per-step information efficiency rather than semantic aggregation. Finally, since MTE is a real-valued front-end executed before spiking computation, we report its dense MACs separately and include them in the overall computation when comparing end-to-end cost; detailed discussion and ablations are deferred to Appendix B.

4.3. Sparsity-Regularized Transform Encoding (STE)

Beyond improving per-step input efficiency, SNNs also require controllable internal activity to preserve sparsity and efficiency. In practice, spiking backbones may compensate for limited horizons by over-firing, which degrades spike-based energy proxies. We therefore propose STE, a plug-and-play transform-encoding neuron module that can replace a controllable fraction p of LIF units and explicitly regularizes firing activity while maintaining accuracy.

Consider a spiking layer receiving multi-step activations $X \in \mathbb{R}^{T \times B \times C \times \Omega}$, where Ω denotes the support ($\Omega = L$ for 1D or $H \times W$ for 2D). STE applies a learnable analysis transform $\mathcal{A}$ to produce latent activations:

$$Y = \mathcal{A}(X), \quad Y \in \mathbb{R}^{T \times B \times K \times \Omega'} \quad (15)$$

Then Eq. (1) uses a multi-spiking neuron $\phi(\cdot)$ to obtain integer pulse codes:

$$S = \phi(Y), \quad S \in \{0, 1, \dots, m\}^{T \times B \times K \times \Omega'} \quad (16)$$

where m is the maximum spikes per step. We set $K = C$ by default to preserve the layer interface; when $K \neq C$, we use a lightweight 1×1 projection to match channel dimensions. The codes S are then fed to subsequent spiking layers, keeping the backbone spike-driven.

To encourage information-preserving yet sparse codes, STE includes a *synthesis* transform $\mathcal{B}$ for reconstruction and is trained with an auxiliary objective.

$$\hat{X} = \mathcal{B}(S), \quad \hat{X} \in \mathbb{R}^{T \times B \times C \times \Omega} \quad (17)$$

$$\mathcal{L}_{\text{STE}} \triangleq \lambda_{\text{rec}} \|X - \hat{X}\|_2^2 + \lambda_{\text{sp}} \|S\|_1 \quad (18)$$

where the sparsity term directly regularizes event counts and thus reduces spike-based energy proxies (Section 3).

Crucially, the synthesis branch $\mathcal{B}$ and the reconstruction loss are used *only during training*; at inference, we remove $\mathcal{B}$ and keep only the forward path $X \xrightarrow{\mathcal{A}} Y \xrightarrow{\phi} S$, so STE does not introduce inference-time reconstruction overhead.

STE is plug-and-play: for a backbone with LIF units, we replace a controllable fraction $p \in [0, 1]$ (evaluated at $p \in \{0, 0.25, 0.5, 0.75, 1\}$), providing a direct accuracy-energy trade-off knob; we further ablate T and the latent width K to trace the Pareto frontier. Its integer spike codes can be treated as event counts (equivalently, virtual binary spikes; see Appendix B), and the sparsity term in Eq. (18) directly regularizes these event counts. At inference, the synthesis branch is removed, and only the analysis transform remains.

4.4. Overall Framework and Training Objective

We integrate MTE and STE into a generic SNN classifier in a plug-and-play manner. Given an input segment $X \in \mathbb{R}^{B \times C \times L}$, MTE produces a T -step real-valued drive stream $\{X^{(t)}\}_{t=1}^T$ (Section 4.2), which is fed to a spiking backbone. Inside the backbone, a controllable fraction p of target LIF units are replaced by STE modules (Section 4.3); the remaining units keep standard LIF dynamics. The network aggregates multi-step outputs using a standard rate/sum readout and optimizes the classification loss jointly with STE’s auxiliary objectives.

Let $\mathcal{S}(p)$ denote the set of layers (or neuron groups) where STE is activated under replacement ratio p . For each $\ell \in \mathcal{S}(p)$, STE produces spike codes S_ℓ and (training-only) reconstructions $\hat{X}_\ell$ with the auxiliary loss in Eq. (18). The overall training objective is:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{STE}} \sum_{\ell \in \mathcal{S}(p)} \left(\lambda_{\text{rec}} \|X_\ell - \hat{X}_\ell\|_2^2 + \lambda_{\text{sp}} \|S_\ell\|_1 \right) \quad (19)$$

where $\mathcal{L}_{\text{cls}}$ is the standard cross-entropy loss and λ_{STE} controls the contribution of STE regularization. At inference, we remove all synthesis branches and reconstructions, keeping only the forward paths that output spiking codes, so there is no inference-time reconstruction overhead.

5. Experiments

5.1. Experimental Setup

Datasets. We evaluate on ten continuous 1D temporal signal classification benchmarks spanning six domains: modulation (RML2016a/b), fault diagnosis (CWRU4/10), ECG (MIT-BIH, PTB-XL), EMG (Ninapro_E1/E2), speech (SCD), and EEG (SEED). We follow the standard protocol of each dataset with a 7:2:1 train/val/test split unless stated otherwise. More details are in Appendix C.1.

Backbones and baselines. We cover four backbone fam-

Algorithm 1 Training with MTE×STE SNN classification

Require: Mini-batch (X, y) , horizon T , backbone f_θ , MTE encoder MTE, STE set $\mathcal{S}(p)$, weights $\lambda_{\text{STE}}, \lambda_{\text{rec}}, \lambda_{\text{sp}}$

- 1: $\{X^{(t)}\}_{t=1}^T \leftarrow \text{MTE}(X)$ (real-valued drive stream)
- 2: Initialize SNN states (membrane, synapses) in f_θ
- 3: **for** $t = 1$ to T **do**
- 4: Forward one step: $(o_t, \{(X_\ell, \hat{X}_\ell, S_\ell)\}_{\ell \in \mathcal{S}(p)}) \leftarrow f_\theta(X^{(t)})$
- 5: **end for**
- 6: Readout: $\hat{y} \leftarrow \text{Readout}(\{o_t\}_{t=1}^T)$ (rate/sum)
- 7: $\mathcal{L}_{\text{cls}} \leftarrow \text{CE}(\hat{y}, y)$
- 8: $\mathcal{L}_{\text{aux}} \leftarrow \sum_{\ell \in \mathcal{S}(p)} (\lambda_{\text{rec}} \|X_\ell - \hat{X}_\ell\|_2^2 + \lambda_{\text{sp}} \|S_\ell\|_1)$
- 9: $\mathcal{L} \leftarrow \mathcal{L}_{\text{cls}} + \lambda_{\text{STE}} \times \mathcal{L}_{\text{aux}}$
- 10: Update θ with surrogate-gradient BPTT on $\mathcal{L}$

ilies: RNN/LSTM, TCN, 1D CNNs (LeNet/ResNet), and Transformer models. For each backbone, we compare its ANN counterpart, a directly trained SNN baseline, and Ours under the matched-backbone/matched-budget protocol (Section 3), i.e., identical macro-architecture and comparable optimization budgets without extra tricks. Specifically, all methods use the same train/validation/test split, data pre-processing, optimizer family, number of training epochs, early-stopping rule, batch size, and per-run training budget. To avoid an additional model-selection advantage, the STE replacement ratio is selected once from a separate backbone-level replacement study and then fixed for the main comparisons, rather than being re-tuned for each dataset, backbone, or baseline comparison. Due to heterogeneous SOTA definitions across 1D benchmarks, we use matched ANNs as task-wise references and directly trained SNNs as standard spiking baselines for controlled comparisons. Model configurations are listed in Appendix C.2.

Implementation and metrics. All models are implemented in SpikingJelly (Fang et al., 2023). We report ACC and use SR as spike-based efficiency proxies (Section 3). Unless stated otherwise, SNNs use a short horizon $T=4$, and we ablate T , STE replacement ratio p , and transform width K in Section 5.3. Training details and hyperparameters are provided in Appendix C.3.

5.2. Main Results

Table 1 reports results on 10 heterogeneous continuous 1D temporal signal benchmarks spanning six task domains under a matched-backbone, matched-budget protocol. Two aggregate trends emerge from the backbone/dataset averages. First, under a short horizon ($T=4$), the performance gap between matched ANNs and directly trained SNN baselines on *native continuous 1D temporal signals* is already modest: averaged over datasets and backbones, SNN attains

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Table 1. Main results on 10 heterogeneous continuous 1D temporal signal benchmarks under matched-backbone/matched-budget protocol. We report ACC / Rating (lower is better); red marks ACC surpassing the matched ANN, and blue marks Rating lower than the matched SNN baseline. Additional metrics (F1) and extended results are provided in the appendix D.1.

Backbone	Variant	Modulation		Fault diagnosis		ECG		EMG		Speech	EEG	Avg. (10 DS)
		RML2016a	RML2016b	CWRU4	CWRU10	MIT-BIH	PTB-XL	Nina-E1	Nina-E2	SCD	SEED	
RNN	ANN	.374 / –	.458 / –	.731 / –	.693 / –	.995 / –	.678 / –	.735 / –	.715 / –	.725 / –	.896 / –	.700 / –
	SNN	.369 / .160	.424 / .248	.964 / .151	.904 / .158	.995 / .309	.677 / .269	.600 / .233	.643 / .274	.793 / .205	.881 / .320	.725 / .233
	Ours	.447 / .174	.487 / .227	.983 / .188	.921 / .082	.994 / .204	.685 / .082	.812 / .280	.815 / .407	.813 / .141	.891 / .306	.785 / .209
LSTM	ANN	.395 / –	.479 / –	.947 / –	.896 / –	.995 / –	.682 / –	.720 / –	.734 / –	.777 / –	.904 / –	.753 / –
	SNN	.375 / .193	.446 / .252	.992 / .051	.832 / .152	.994 / .213	.661 / .344	.566 / .563	.606 / .593	.426 / .131	.896 / .553	.679 / .304
	Ours	.429 / .229	.482 / .274	.994 / .172	.931 / .206	.994 / .180	.670 / .249	.738 / .492	.671 / .552	.526 / .110	.899 / .530	.733 / .299
TCN	ANN	.602 / –	.637 / –	.914 / –	.846 / –	.956 / –	.521 / –	.667 / –	.630 / –	.908 / –	.754 / –	.744 / –
	SNN	.569 / .168	.596 / .205	.878 / .184	.846 / .167	.950 / .223	.515 / .171	.530 / .145	.549 / .144	.885 / .151	.674 / .189	.699 / .175
	Ours	.608 / .112	.647 / .087	.964 / .087	.904 / .103	.956 / .088	.554 / .085	.591 / .095	.592 / .111	.912 / .077	.800 / .105	.753 / .095
LeNet5-1D	ANN	.492 / –	.521 / –	.979 / –	.996 / –	.990 / –	.660 / –	.625 / –	.645 / –	.811 / –	.770 / –	.749 / –
	SNN	.428 / .272	.472 / .343	.959 / .239	.992 / .296	.993 / .290	.649 / .218	.496 / .302	.545 / .329	.712 / .321	.704 / .216	.695 / .283
	Ours	.465 / .152	.496 / .156	.989 / .165	.997 / .166	.996 / .142	.653 / .125	.504 / .156	.588 / .182	.717 / .159	.770 / .124	.718 / .153
ResNet18-1D	ANN	.576 / –	.639 / –	.996 / –	.966 / –	.994 / –	.705 / –	.900 / –	.896 / –	.932 / –	.815 / –	.842 / –
	SNN	.582 / .255	.623 / .250	.975 / .227	.960 / .238	.995 / .232	.694 / .219	.882 / .233	.847 / .236	.925 / .252	.778 / .235	.826 / .238
	Ours	.564 / .178	.626 / .165	.999 / .240	.968 / .204	.997 / .138	.706 / .156	.903 / .188	.900 / .206	.934 / .161	.919 / .207	.852 / .184
Spikformer-1D	ANN	.550 / –	.605 / –	.997 / –	.970 / –	.995 / –	.672 / –	.862 / –	.870 / –	.840 / –	.859 / –	.822 / –
	SNN	.486 / .129	.560 / .172	.997 / .088	.976 / .141	.995 / .125	.672 / .062	.870 / .140	.867 / .165	.817 / .260	.815 / .123	.805 / .141
	Ours	.569 / .121	.591 / .093	.999 / .053	.983 / .094	.997 / .100	.676 / .039	.809 / .091	.833 / .101	.825 / .111	.844 / .085	.813 / .089
Avg. Backbones	ANN	.498 / –	.558 / –	.927 / –	.894 / –	.987 / –	.653 / –	.751 / –	.748 / –	.832 / –	.833 / –	.768 / –
	SNN	.468 / .196	.520 / .245	.961 / .157	.918 / .192	.987 / .232	.645 / .214	.657 / .269	.676 / .290	.760 / .220	.791 / .273	.738 / .229
	Ours	.514 / .164	.550 / .172	.988 / .151	.951 / .142	.989 / .159	.657 / .123	.726 / .217	.733 / .260	.788 / .127	.854 / .234	.775 / .175

0.738 ACC versus 0.768 for ANN (Avg.Backbones). Second, **Ours** further improves both accuracy and efficiency: on Avg.Backbones, Ours reaches 0.775 ACC while reducing Rating from 0.229 (SNN) to 0.175 (Ours), indicating a favorable shift of the accuracy–efficiency frontier rather than an accuracy–efficiency trade-off.

The improvement is consistent across backbone families and task domains. For each backbone, Ours improves over its SNN counterpart on the per-backbone mean over 10 datasets, e.g., RNN: 0.785 vs. 0.725, TCN: 0.753 vs. 0.699, and ResNet18-1D: 0.852 vs. 0.826, while achieving lower Rating in all cases. Across domains, the Avg.Backbones row shows gains in modulation, fault diagnosis, ECG/EMG, and speech/EEG benchmarks, suggesting broad applicability under this controlled evaluation protocol.

These trends are consistent with the mechanism in Section 4.1. With limited timesteps, accuracy hinges on increasing per-step temporal evidence while avoiding redundant firing that wastes the spiking budget. MTE addresses the former by replacing naive repetition with a multi-scale temporal stream, whereas STE addresses the latter via a sparsity-regularized transform-encoding bottleneck. Accordingly, Table 1 shows simultaneous ACC improvements and Rating reductions, matching the proposed “densify-then-compress” principle.

The magnitude of the gains also follows this interpretation.

MTE×STE tends to help most when discriminative information appears as local or multi-scale temporal evidence, where MTE can diversify short-horizon inputs and STE can remove redundant firing. By contrast, the gains can be weaker or less stable in deeper residual or Transformer-style backbones, where stronger spatiotemporal mixing may make a fixed STE replacement ratio suboptimal. This suggests that the current fixed replacement policy is a conservative design for controlled comparison, but not a fully adaptive solution; learning layer-wise or input-dependent replacement and sparsity policies remains an important direction for future work. Extended results with ACC/F1/Rating and further ablations are provided in Appendix D.1.

5.3. Ablation Studies

We ablate components and hyper-parameters to study the accuracy-activity trade-off under short-horizon inference, which is the regime analyzed in Section 4.1. We report ACC and spike rate in the main paper and defer full results (F1 and extended sweeps) to the appendix. Unless specified, we use $T=4$ and the default MTE schedule.

5.3.1. COMPONENT ABLATION

Table 2 isolates the contributions of MTE and STE on the SLSTM backbone. MTE improves short-horizon temporal evidence but generally increases spike activity, whereas

Table 2. Component ablation on the SLSTM backbone over three representative datasets. ACC denotes accuracy and R denotes the average spike rate, where lower R indicates sparser activity.

Variant	SEED		MIT-BIH		CWRU10		AVG	
	ACC	R	ACC	R	ACC	R	ACC	R
ANN	.904	—	.995	—	.896	—	.932	—
SNN	.896	.553	.994	.213	.832	.152	.907	.306
STE	.951	.439	.995	.207	.851	.133	.932	.260
MTE	.904	.619	.994	.290	.911	.234	.936	.381
Ours	.899	.530	.994	.180	.931	.206	.941	.305

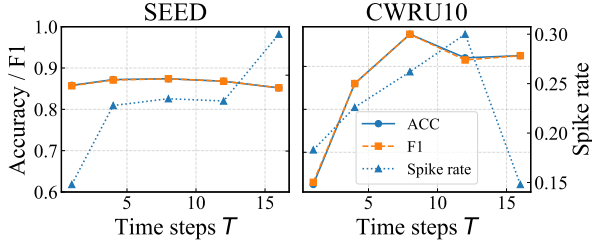


Figure 3. MTE horizon T sweep: ACC and spike rate versus T on SLSTM for SEED and CWRU10.

STE regularizes firing via sparse transform coding and can recover accuracy with controlled activity. Crucially, combining them (MTE + STE) yields the most favorable accuracy-activity trade-off, supporting the complementary design of MTE $\times$ STE. More results on additional backbones and full metrics are reported in Appendix D.2.

5.3.2. MTE ANALYSIS

MTE T sweep. We vary the simulation horizon T used by MTE and report accuracy and spike activity. Figure 3 shows representative results on the SLSTM backbone for SEED and CWRU10. Across datasets, performance depends non-monotonically on T : too small T may under-utilize temporal evidence, while overly large T tends to increase spike activity with diminishing returns. A moderate horizon yields a favorable accuracy-activity balance, with task-dependent optima (e.g., SEED peaks around $T=8$ whereas CWRU10 prefers a slightly longer context). This behavior is consistent with the variance accumulation analysis in Section 4.1, where increasing T beyond a moderate range primarily amplifies encoding and integration noise. Complete sweeps across additional datasets and the SLeNet backbone are provided in Appendix D.3.

Multi-bit allocation schedules. We further ablate integer-spike allocation schedules on the SLSTM backbone (Table 3). Results on SLeNet and full metrics are deferred to Appendix D.4. Unlike a naive pyramid that only changes resolution, these schedules explicitly control the discrete precision budget per scale. Overall, scale-aligned allocations

Table 3. Multi-bit schedule ablation for MTE on SLSTM: We report ACC and activity rate (R) under different integer-spike allocation schedules.

Dataset	1111		1234		4321		4444	
	ACC	R	ACC	R	ACC	R	ACC	R
CWRU4	.994	.053	.992	.061	.992	.058	.994	.054
SEED	.867	.656	.852	.634	.859	.674	.867	.585
PTB-XL	.659	.375	.670	.386	.674	.393	.658	.361
Nina-E1	.588	.600	.569	.614	.589	.632	.581	.645
Nina-E2	.589	.648	.589	.671	.578	.629	.585	.623

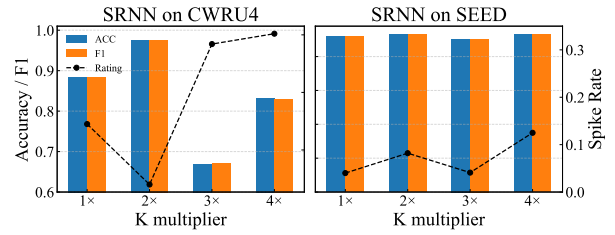


Figure 4. STE dictionary width (K) sweep. Accuracy and spike rating versus K on SRNN for representative datasets.

yield robust accuracy-rate trade-offs under short horizons; among them, 4321 provides a consistently strong balance and is used as the default. Full results (including F1 and the SLeNet backbone) are deferred to Appendix D.4.

5.3.3. STE ANALYSIS

We ablate the STE dictionary width K , reported as multiples of the baseline channel size. Fig. 4 shows representative results on SRNN, more results across all datasets and backbones are provided in Appendix D.5. Varying K reveals a consistent capacity-activity trade-off: small K may under-fit due to limited coding capacity, whereas overly large K introduces redundant degrees of freedom and substantially increases spike activity. Across tasks, a moderate K typically achieves the best balance, yielding higher accuracy with lower or comparable activity. This observation motivates the default K used throughout the paper. Compared to the horizon sweep in MTE, varying K mainly affects representational redundancy rather than temporal coverage, highlighting the orthogonality of the two components.

5.4. Energy Reduction

We evaluate inference efficiency using SOPs and spike-rate-based accounting as standard algorithm-level proxies; definitions and conversion coefficients are provided in Appendix E. Table 4 reports the inference-time operation and energy proxies under the default short horizon. Directly trained SNN baselines already show lower estimated proxies than ANNs, e.g., SLSTM $69.4 \rightarrow 4.15 \mu\text{J}$ and SLeNet

Table 4. Theoretical power consumption under inference.

Backbone	Variant	$ \theta $ (M)	R	SOPs (M)	$\hat{E}$ (uJ)	$\hat{E}(\Delta)$
SLSTM	ANN	3.768	—	—	69.4	16.7
	SNN	3.770	0.306	4.615	4.15	1.00
	+STE	3.770	0.353	5.324	4.79	1.15
	+MTE	4.029	0.448	7.221	6.50	1.57
	Ours	4.029	0.355	5.722	5.15	1.24
SLeNet	ANN	0.022	—	—	3.71	30.7
	SNN	0.022	0.267	0.023	0.121	1.00
	+STE	0.021	0.130	0.011	0.111	0.92
	+MTE	0.281	0.245	0.275	0.247	2.04
	Ours	0.281	0.144	0.162	0.146	1.21

$3.71 \rightarrow 0.121 \mu\text{J}$. MTE enriches multi-scale temporal evidence but increases activity and SOPs, while STE partially compresses this redundancy, reducing SOPs from MTE-only to MTE $\times$ STE, e.g., SLSTM $7.221 \rightarrow 5.722\text{M}$ and SLeNet $0.275 \rightarrow 0.162\text{M}$. STE alone may increase SOPs, as in SLSTM+STE, because the analysis transform can introduce additional latent activity without preceding multi-scale enrichment. Thus, the results support the proposed densify-then-compress design, but SOPs/SR and converted μJ values should be interpreted as relative algorithm-level efficiency indicators rather than direct measurements of wall-clock latency or chip-level power. Exact layer-wise SOPs accounting is reported in Appendix E.

6. Conclusion

This paper revisits SNNs for continuous 1D temporal signal classification from a phenomenon- and mechanism-driven perspective. Under matched architectures and training budgets, the ANN-SNN gap is already modest on native temporal signals at short horizons, consistent with leaky integration as causal memory. Guided by this, we propose a plug-and-play MTE $\times$ STE framework that enriches short-horizon evidence while suppressing redundant firing. Across ten 1D benchmarks and multiple backbone families, it consistently improves the accuracy-efficiency trade-off over standard SNNs and matches or exceeds ANNs in several settings. This study is limited to continuous sensory 1D signals, relies on SOP/SR-based algorithmic efficiency proxies rather than chip-level measurements, and compares mainly with matched full-precision ANNs; adaptive STE replacement policies and comparisons with compressed ANN variants remain important future directions. We hope this encourages task-aligned evaluation and principled short-horizon spiking designs.

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Impact Statement

This paper presents work whose primary goal is to advance the understanding and design of spiking neural networks for efficient one-dimensional signal classification. The proposed methods focus on representation efficiency, energy-aware computation, and principled comparisons under matched experimental settings. We do not foresee any immediate negative societal or ethical consequences arising directly from this work. As with many advances in machine learning, downstream impacts depend on specific application contexts, and we encourage responsible use consistent with established best practices.

References

- Amir, A., Taba, B., Berg, D., Melano, T., McKinstry, J., Di Nolfo, C., Nayak, T., Andreopoulos, A., Garreau, G., Mendoza, M., et al. A low power, fully event-based gesture recognition system. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 7243–7252, 2017.
- Bai, S., Kolter, J. Z., and Koltun, V. An empirical evaluation of generic convolutional and recurrent networks for sequence modeling. *arXiv preprint arXiv:1803.01271*, 2018.
- Bi, J., Wang, F., Ping, J., Qu, G., Hu, F., Li, H., and Han, S. Fbn-tcn: Temporal convolutional neural network based on spatial domain fusion brain networks for affective brain-computer interfaces. *Biomedical Signal Processing and Control*, 94:106323, 2024.
- Carbajal, D. and Romero, J. L. Model agnostic signal encoding by leaky integrate and fire, performance and uncertainty. *arXiv*, Jan 2024. doi: 10.48550/ARXIV.2412.12994. URL <https://arxiv.org/abs/2412.12994>.
- Dayan, P. and Abbott, L. F. *Theoretical neuroscience: computational and mathematical modeling of neural systems*. MIT press, 2005.
- Ding, J., Yu, Z., Tian, Y., and Huang, T. Optimal ann-snn conversion for fast and accurate inference in deep spiking neural networks. *arXiv preprint arXiv:2105.11654*, 2021.

- Dominguez-Morales, J. P., Liu, Q., James, R., Gutierrez-Galan, D., Jimenez-Fernandez, A., Davidson, S., and Furber, S. Deep spiking neural network model for time-variant signals classification: a real-time speech recognition approach. In *2018 International Joint Conference on Neural Networks (IJCNN)*, pp. 1–8. IEEE, 2018.
- Duan, R.-N., Zhu, J.-Y., and Lu, B.-L. Differential entropy feature for eeg-based emotion classification. In *2013 6th international IEEE/EMBS conference on neural engineering (NER)*, pp. 81–84. IEEE, 2013.
- El Bouny, L., Khalil, M., and Adib, A. Ecg heartbeat classification based on multi-scale wavelet convolutional neural networks. In *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 3212–3216. IEEE, 2020.
- Fang, H., Shrestha, A., and Qiu, Q. Multivariate time series classification using spiking neural networks. In *2020 International Joint Conference on Neural Networks (IJCNN)*, pp. 1–7. IEEE, 2020.
- Fang, W., Yu, Z., Chen, Y., Huang, T., Masquelier, T., and Tian, Y. Deep residual learning in spiking neural networks. *Advances in Neural Information Processing Systems*, 34:21056–21069, 2021.
- Fang, W., Chen, Y., Ding, J., Yu, Z., Masquelier, T., Chen, D., Huang, L., Zhou, H., Li, G., and Tian, Y. Spikingjelly: An open-source machine learning infrastructure platform for spike-based intelligence. *Science Advances*, 9(40): eadi1480, 2023.
- Feng, L., Liu, Q., Tang, H., Ma, D., and Pan, G. Multi-level firing with spiking ds-resnet: Enabling better and deeper directly-trained spiking neural networks. *arXiv preprint arXiv:2210.06386*, 2022.
- Gerstner, W. and Kistler, W. M. *Spiking neuron models: Single neurons, populations, plasticity*. Cambridge university press, 2002.
- He, K., Zhang, X., Ren, S., and Sun, J. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 770–778, 2016.
- Hochreiter, S. and Schmidhuber, J. Long short-term memory. *Neural computation*, 9(8):1735–1780, 1997.
- Langley, P. Crafting papers on machine learning. In Langley, P. (ed.), *Proceedings of the 17th International Conference on Machine Learning (ICML 2000)*, pp. 1207–1216, Stanford, CA, 2000. Morgan Kaufmann.
- LeCun, Y., Bottou, L., Bengio, Y., and Haffner, P. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.
- Li, H., Liu, H., Ji, X., Li, G., and Shi, L. Cifar10-dvs: an event-stream dataset for object classification. *Frontiers in neuroscience*, 11:309, 2017.
- Lv, C., Xu, J., and Zheng, X. Spiking convolutional neural networks for text classification. *arXiv preprint arXiv:2406.19230*, 2024.
- Maass, W. Networks of spiking neurons: the third generation of neural network models. *Neural networks*, 10(9): 1659–1671, 1997.
- Moody, G. B. and Mark, R. G. The impact of the mit-bih arrhythmia database. *IEEE engineering in medicine and biology magazine*, 20(3):45–50, 2001.
- O’shea, T. J. and West, N. Radio machine learning dataset generation with gnu radio. In *Proceedings of the GNU radio conference*, volume 1, 2016.
- O’Shea, T. J., Corgan, J., and Clancy, T. C. Convolutional radio modulation recognition networks. In *Engineering Applications of Neural Networks: 17th International Conference, EANN 2016, Aberdeen, UK, September 2-5, 2016, Proceedings 17*, pp. 213–226. Springer, 2016.
- Phyo, J., Ko, W., Jeon, E., and Suk, H.-I. Transsleep: Transitioning-aware attention-based deep neural network for sleep staging. *IEEE Transactions on Cybernetics*, 53(7):4500–4510, 2022.
- Prabhakararao, E. and Dandapat, S. Multi-scale convolutional neural network ensemble for multi-class arrhythmia classification. *IEEE Journal of Biomedical and Health Informatics*, 26(8):3802–3812, 2021.
- Sengupta, A., Ye, Y., Wang, R., Liu, C., and Roy, K. Going deeper in spiking neural networks: Vgg and residual architectures. *Frontiers in neuroscience*, 13:95, 2019.
- Smith, W. A. and Randall, R. B. Rolling element bearing diagnostics using the case western reserve university data: A benchmark study. *Mechanical systems and signal processing*, 64:100–131, 2015.
- Song, T., Zheng, W., Song, P., and Cui, Z. Eeg emotion recognition using dynamical graph convolutional neural networks. *IEEE Transactions on Affective Computing*, 11(3):532–541, 2018.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., and Polosukhin, I. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.
- Wagner, P., Strodthoff, N., Bousseljot, R.-D., Kreiseler, D., Lunze, F. I., Samek, W., and Schaeffter, T. Ptb-xl, a large publicly available electrocardiography dataset. *Scientific data*, 7(1):1–15, 2020.

- Wang, Z., Shangguan, W., Peng, C., and Cai, B. A fault diagnosis method based on data feature reconstruction and deep transfer learning. In *2022 IEEE Asia-Pacific Conference on Image Processing, Electronics and Computers (IPEC)*, pp. 1–5. IEEE, 2022.
- Wu, Y., Deng, L., Li, G., Zhu, J., Xie, Y., and Shi, L. Direct training for spiking neural networks: Faster, larger, better. In *Proceedings of the AAAI conference on artificial intelligence*, volume 33, pp. 1311–1318, 2019.
- Xu, H., Gu, Z., Sun, R., and Ma, D. Training multi-bit spiking neural network with virtual neurons. *Neuro-computing*, 634:129825, 2025. ISSN 0925-2312. doi: <https://doi.org/10.1016/j.neucom.2025.129825>.
- Yan, Z., Tang, K., Zhou, J., and Wong, W.-F. Low latency conversion of artificial neural network models to rate-encoded spiking neural networks. *IEEE Transactions on Neural Networks and Learning Systems*, 36(8):14107–14118, 2025.
- Zhang, A., Li, S., Cui, Y., Yang, W., Dong, R., and Hu, J. Limited data rolling bearing fault diagnosis with few-shot learning. *Ieee Access*, 7:110895–110904, 2019.
- Zhang, J., Wang, T., Feng, Z., and Yang, S. Toward the automatic modulation classification with adaptive wavelet network. *IEEE Transactions on Cognitive Communications and Networking*, 9(3):549–563, 2023.
- Zheng, H., Wu, Y., Deng, L., Hu, Y., and Li, G. Going deeper with directly-trained larger spiking neural networks. In *Proceedings of the AAAI conference on artificial intelligence*, volume 35, pp. 11062–11070, 2021.
- Zheng, W.-L. and Lu, B.-L. Investigating critical frequency bands and channels for eeg-based emotion recognition with deep neural networks. *IEEE Transactions on autonomous mental development*, 7(3):162–175, 2015.
- Zhou, Z., Zhu, Y., He, C., Wang, Y., Yan, S., Tian, Y., and Yuan, L. Spikformer: When spiking neural network meets transformer. *arXiv preprint arXiv:2209.15425*, 2022.

A. Detailed Derivations and Additional Analysis

This appendix complements Section 4.1 by (i) relaxing the temporal independence used in Proposition 1, (ii) contrasting stochastic vs. deterministic static-to-spike temporalization under a finite horizon, (iii) discussing reset and spiking nonlinearity via standard local surrogate arguments, and (iv) linking state/logit uncertainty to conservative decision stability bounds under short horizons. Unless otherwise stated, we analyze a single projected coordinate (or a single neuron state); the statements hold element-wise and extend to vector states by applying the bounds coordinate-wise (or via trace bounds).

A.1. Proof of Proposition 1 and Correlated-noise Extension

Recall the linearized dynamics (Eq. (5)) and decomposition (Eq. (7)):

$$u_t = \alpha u_{t-1} + I_t, \quad \alpha \in (0, 1), \quad u_0 = 0 \quad (20)$$

$$I_t = \bar{I}_t + \epsilon_t^{(\text{in})} + \epsilon_t^{(\text{enc})} \quad (21)$$

Let $\bar{u}_t$ be the state driven by $\bar{I}_t$ and define $\Delta u_t \triangleq u_t - \bar{u}_t$. Expanding recursively yields:

$$\Delta u_T = \sum_{k=1}^T \alpha^{T-k} \left(\epsilon_k^{(\text{in})} + \epsilon_k^{(\text{enc})} \right) \quad (22)$$

Define $w_k \triangleq \alpha^{T-k}$ and $\eta_k \triangleq \epsilon_k^{(\text{in})} + \epsilon_k^{(\text{enc})}$, so $\Delta u_T = \sum_{k=1}^T w_k \eta_k$ and Eq. (23).

$$\text{Var}(\Delta u_T) = \sum_{i=1}^T \sum_{j=1}^T w_i w_j \text{Cov}(\eta_i, \eta_j). \quad (23)$$

A general bound under temporal dependence. Assume $\mathbb{E}[\eta_t] = 0$ and define an absolute autocovariance envelope:

$$\Gamma(h) \triangleq \sup_t |\text{Cov}(\eta_t, \eta_{t+h})|, \quad h \geq 0 \quad (24)$$

By the triangle inequality and Eq. (23), we can conclude that:

$$\begin{aligned} \text{Var}(\Delta u_T) &\leq \sum_{i=1}^T \sum_{j=1}^T w_i w_j \Gamma(|i-j|) \\ &= \sum_{h=0}^{T-1} \Gamma(h) \sum_{j=1}^{T-h} w_{j+h} w_j \end{aligned} \quad (25)$$

Since $w_k = \alpha^{T-k} \geq 0$, the inner sum admits a closed form:

$$\sum_{j=1}^{T-h} w_{j+h} w_j = \sum_{j=1}^{T-h} \alpha^{(T-(j+h))} \alpha^{(T-j)} \quad (26)$$

$$= \alpha^h \sum_{n=0}^{T-h-1} \alpha^{2n} \leq \alpha^h \sum_{n=0}^{T-1} \alpha^{2n} \quad (27)$$

Combining Eq. (25)–(27), we can obtain:

$$\text{Var}(\Delta u_T) \leq \left(\sum_{n=0}^{T-1} \alpha^{2n} \right) \left(\Gamma(0) + 2 \sum_{h=1}^{T-1} \alpha^h \Gamma(h) \right) \quad (28)$$

Eq. (28) shows that exponentially decaying memory weights suppress long-lag dependence in the disturbance as long as the autocovariance does not grow too slowly; this supports the main-text statement that strict temporal independence is not essential for the qualitative comparison in Proposition 1.

Separating input and encoding disturbances without independence. Writing $\eta_t = \epsilon_t^{(\text{in})} + \epsilon_t^{(\text{enc})}$, Eq. (28) holds directly for the total disturbance. If one wishes to make sources explicit, an upper bound follows from envelopes of each term and their cross-covariances:

$$\Gamma(h) \leq \Gamma_{\text{in}}(h) + \Gamma_{\text{enc}}(h) + 2\Gamma_{\text{cross}}(h) \quad (29)$$

where $\Gamma_{\text{in}}(h) \triangleq \sup_t |\text{Cov}(\epsilon_t^{(\text{in})}, \epsilon_{t+h}^{(\text{in})})|$, etc. This relaxation justifies that the clean additive variance form used in the main text is a convenient special case, not a requirement.

A.2. Static-to-spike temporalization: stochastic vs. deterministic finite- T error

Section 4.1 models temporalization by an additional encoding-induced term $\epsilon^{(\text{enc})}$ in the drive I_t . Here we illustrate two common regimes under a finite horizon.

(i) Stochastic rate-like encoding (Bernoulli/Poisson). Given a static scalar amplitude a (after normalization), rate-like coding generates a spike-like drive $\{I_t\}$ over T steps with $\mathbb{E}[I_t] = a$ (or proportional to a):

- **Bernoulli coding:** $I_t \sim \text{Bernoulli}(a)$ so $\text{Var}(I_t) = a(1-a)$.
- **Poisson coding:** $I_t \sim \text{Poisson}(\lambda)$ with $\lambda \propto a$ so $\text{Var}(I_t) = \lambda$.

Under conditional independence across t and using $\Delta I_t \triangleq I_t - \mathbb{E}[I_t]$, Eq. (22) gives

$$\text{Var}(\Delta u_T) = \sum_{t=1}^T \alpha^{2(T-t)} \text{Var}(\Delta I_t) \quad (30)$$

$$\leq \left(\sup_t \text{Var}(\Delta I_t) \right) \sum_{n=0}^{T-1} \alpha^{2n} \quad (31)$$

With temporal dependence, Eq. (28) provides the corresponding bound by replacing $\Gamma(h)$ with the autocovariance envelope of ΔI_t .

(ii) Deterministic temporal encoding and finite-resolution error. Many short-horizon pipelines use deterministic temporal encoders (e.g., latency / time-to-first-spike / rank-order coding) where a static amplitude is mapped to discrete time indices due to finite T and discrete time steps. In this case, uncertainty manifests as a *finite-resolution error* rather than stochastic variance.

A generic model is to assume a bounded per-step encoding deviation:

$$I_t = \tilde{I}_t + e_t^{(\text{enc})}, \quad \|e_t^{(\text{enc})}\| \leq \varepsilon_T \quad (32)$$

where $\tilde{I}_t$ denotes the (encoder-dependent) target drive at horizon T and ε_T captures discretization/quantization induced by finite T . Then from Eq. (22) and triangle inequality,

$$|\Delta u_T| = \left| \sum_{k=1}^T \alpha^{T-k} e_k^{(\text{enc})} \right| \quad (33)$$

$$\leq \sum_{k=1}^T \alpha^{T-k} |e_k^{(\text{enc})}| \leq \varepsilon_T \frac{1 - \alpha^T}{1 - \alpha} \quad (34)$$

Eq. (34) formalizes an irreducible finite- T error floor for deterministic temporalization unless ε_T is reduced (e.g., by richer per-step representational capacity).

A.3. Multi-bit / multi-spike coding under short horizons

Our method uses integer-valued (multi-spike) neurons to allocate scale-aligned precision across temporal scales (Section 4.2), which can be interpreted as increasing per-step representational capacity under fixed small T .

A generic multi-bit model. Let $I_t \in \{0, 1, \dots, M\}$ be an integer-valued drive at each step with $\mathbb{E}[I_t] = \mu$ and $\text{Var}(I_t) = \sigma^2$. Under conditional independence across time:

$$\mathbb{E}[u_T] = \mu \sum_{n=0}^{T-1} \alpha^n, \quad \text{Var}(\Delta u_T) = \sigma^2 \sum_{n=0}^{T-1} \alpha^{2n} \quad (35)$$

Eq. (35) matches the stochastic bound structure in Eq. (31), with σ^2 determined by the coding rule. For deterministic temporalization, increasing per-step levels can be viewed as reducing the effective resolution parameter ε_T in Eq. (32) at fixed small T .

Correlated multi-bit streams. If I_t is temporally correlated, Eq. (28) applies by letting $\eta_t = I_t - \mu$ and using the corresponding $\Gamma(h)$.

A.4. Incorporating reset: boundedness and perturbation view

The full discrete-time update (Eq. (1)) includes a reset term, written generically as

$$u_t = \alpha u_{t-1} + I_t - \vartheta s_{t-1} \quad (36)$$

If spikes are bounded by design (binary spikes or clipped multi-spike outputs), i.e., $s_t \in [0, s_{\max}]$, then the reset contribution is bounded:

$$|-\vartheta s_{t-1}| \leq \vartheta s_{\max} \quad (37)$$

Reset introduces a state-dependent, bounded term that may change constants and coupling, but the key distinction highlighted by Proposition 1 remains: in static temporalization an additional encoder-induced perturbation enters the drive I_t , and this perturbation propagates through the same leaky accumulation mechanism (now with reset), potentially amplifying state uncertainty under short horizons.

A.5. Spiking nonlinearity and surrogate-based local sensitivity

To connect the linearized view to practical training with surrogate gradients, consider the surrogate approximation $s_t \approx \phi(u_t)$ where $\phi(\cdot)$ is smooth in the operating region and has bounded slope:

$$|\phi'(u)| \leq L_\phi \quad \text{for } u \in \mathcal{U} \quad (38)$$

Consider two trajectories driven by I_t and $\bar{I}_t$ under Eq. (36) with the same parameters, and define $\Delta u_t = u_t - \bar{u}_t$ and $\Delta I_t = I_t - \bar{I}_t$. By the mean value theorem and Eq. (38),

$$|\phi(u_{t-1}) - \phi(\bar{u}_{t-1})| \leq L_\phi |\Delta u_{t-1}| \quad (39)$$

$$|\Delta u_t| \leq \alpha |\Delta u_{t-1}| + |\Delta I_t| + \vartheta L_\phi |\Delta u_{t-1}| \quad (40)$$

$$= (\alpha + \vartheta L_\phi) |\Delta u_{t-1}| + |\Delta I_t| \quad (41)$$

A sufficient *local* stability condition is $\alpha + \vartheta L_\phi < 1$, under which iterating Eq. (41) yields:

$$|\Delta u_T| \leq \sum_{k=1}^T (\alpha + \vartheta L_\phi)^{T-k} |\Delta I_k| \quad (42)$$

Even when the sufficient condition is not strictly met, Eq. (41) still shows that encoder perturbations are modulated by an effective local gain $\alpha_{\text{eff}} = \alpha + \vartheta L_\phi$ in the operating regime, supporting the main-text interpretation that temporalization-induced perturbations can translate into state uncertainty at short horizons.

A.6. From state uncertainty to decision stability (conservative)

Let $g(\cdot)$ denote a scalar logit margin (e.g., the difference between the top-1 and runner-up logits) computed at the readout. Assume g is L_g -Lipschitz with respect to the readout state:

$$|g(u) - g(\bar{u})| \leq L_g \|u - \bar{u}\| \quad (43)$$

Let the clean margin be $m \triangleq g(\bar{u}_T) > 0$ and define $\Delta g \triangleq g(u_T) - g(\bar{u}_T)$. Misclassification occurs if $\Delta g \leq -m$. A conservative bound follows from:

$$\mathbb{P}(\Delta g \leq -m) \leq \mathbb{P}(|\Delta g| \geq m) \leq \frac{\text{Var}(\Delta g)}{m^2} \quad (44)$$

Using Lipschitzness and $\Delta u_T = u_T - \bar{u}_T$:

$$\text{Var}(\Delta g) \leq L_g^2 \text{Var}(\Delta u_T) \quad (45)$$

which together yield:

$$\mathbb{P}(\Delta g \leq -m) \leq \frac{L_g^2 \text{Var}(\Delta u_T)}{m^2} \quad (46)$$

Eq. (46) formalizes that additional encoding-induced uncertainty (increasing $\text{Var}(\Delta u_T)$) disproportionately affects low-margin samples under short horizons, consistent with larger ANN-SNN gaps on static vision when temporalization is required.

A.7. Connection back to MTE and STE

Proposition 1 highlights a short-horizon mismatch: static inputs require temporalization, which introduces an additional encoding-induced perturbation/error component (stochastic variance or finite-resolution error) that propagates through leaky accumulation; native temporal signals can devote the same horizon primarily to causal evidence accumulation.

This motivates two complementary principles:

- **MTE (Section 4.2):** Increase *per-step information efficiency* under fixed small T by allocating representational precision across temporal scales, which can be viewed as reducing the effective per-step temporalization error (e.g., $\sup_t \text{Var}(\Delta I_t)$ in Eq. (31) or ε_T in Eq. (32)) without requiring a longer horizon.
- **STE (Section 4.3):** Preserve the *energy advantage* of spiking by regulating internal firing activity while maintaining discriminative information. Since spike-based energy proxies scale with activity and fan-in (Section E), STE provides a controllable mechanism to suppress redundant activity, mitigating the risk that richer front-end evidence leads to excessive firing.

Together, MTE targets the front-end representational bottleneck under short horizons, while STE targets the internal activity/energy bottleneck, aligning the method with the mechanism-level motivation in Section 4.1.

B. Additional Details on Multi-Scale Temporal Encoding (MTE)

This appendix provides implementation-level details and extended discussions for MTE, including (i) a concrete instantiation of the operators in Eq. (11)–(13), (ii) how multi-bit spikes can be interpreted as event counts (or virtual binary neurons) and how they are accounted for by spike-based proxies, (iii) how we account for non-spiking dense operations in a matched comparison, and (iv) why MTE is conceptually different from feature pyramid networks.

B.1. Instantiation of MTE operators

Scale schedule and boundary cases. Given an input length L , we use the monotone schedule in Eq. (11) with a minimum resolution $L_{\min}$. In practice, to avoid upsampling when $L < L_{\min}$, we set $L_{\min} \leftarrow \min(L_{\min}, L)$.

Direct downsampling and upsampling. For continuous 1D temporal signals, we instantiate $\mathcal{D}_t$ and $\mathcal{U}_t$ with lightweight deterministic resampling:

- **Direct downsampling $\mathcal{D}_t$:** adaptive average pooling (or linear interpolation) from length L to L_t .
- **Upsampling $\mathcal{U}_t$:** linear interpolation from length L_t back to L .

These operators are fixed (non-learned) and serve to enforce scale alignment and stabilize training under short horizons.

Feature refinement branch. Eq. (14) defines $\mathcal{F}_t$ via a lightweight local transform $\mathcal{H}_t$ and a mapping $\mathcal{P}_{t \rightarrow t+1}$. For 1D inputs, we instantiate them as:

$$\mathcal{H}_t(Z_t) = \text{Linear}_t^{(1)}(Z_t) \in \mathbb{R}^{B \times C \times L_{t+1}} \quad (47)$$

$$S_t = \phi_{m_t}(\mathcal{H}_t(Z_t)) \in \{0, 1, \dots, m_t\}^{B \times C \times L_{t+1}} \quad (48)$$

$$\mathcal{P}_{t \rightarrow t+1}(S_t) = \text{Linear}_t^{(2)}(S_t) \in \mathbb{R}^{B \times C \times L_{t+1}} \quad (49)$$

so that $\mathcal{F}_t(Z_t) = \mathcal{P}_{t \rightarrow t+1}(S_t)$ matches the target resolution L_{t+1} in Eq. (12). Here $\text{Linear}_t^{(1)}$ maps the temporal dimension from L_t to L_{t+1} per channel, and $\text{Linear}_t^{(2)}$ refines the resulting integer code into a real-valued drive at the same scale. This realization matches the intent of Eq. (14): the integer code is used *internally* for allocating representational precision, while the backbone always receives a real-valued drive stream $\{X^{(t)}\}$.

Multi-bit schedule. We use a decreasing multi-bit spike budget $\{m_t\}_{t=1}^{T-1}$ (default $[4, 3, 2, 1]$ when $T=4$) so that higher-resolution steps are assigned more spike levels. We ablate T and the spike-budget schedule in Appendix C/Section 5.

B.2. Multi-bit spikes as event counts and spike-based proxies

Event-count interpretation. The multi-spike nonlinearity $\phi_m(\cdot)$ produces an integer-valued code $S \in \{0, 1, \dots, m\}$. Such an integer code can be interpreted as the *count* of binary spike events within a single time step, or equivalently as the sum of spikes emitted by a set of m *virtual* binary neurons:

$$S = \sum_{q=1}^m s^{(q)}, \quad s^{(q)} \in \{0, 1\} \quad (50)$$

This view is standard in multi-level firing SNNs and virtual-neuron training schemes (Feng et al., 2022; Xu et al., 2025). Consequently, multi-bit firing does not inherently require dense multiplications; it corresponds to more (or fewer) events being accumulated.

How SR/SOPs treat multi-bit codes. Our spike-based activity and operation proxies (Section 3) generalize naturally to integer spikes by treating $|S|$ as the event count. Specifically, for an integer-valued spike tensor $S_{t,\ell,i} \in \{0, \dots, m\}$, we use

$$\text{SR} \triangleq \frac{1}{|\Omega|} \sum_{(t,\ell) \in \Omega} |S_{t,\ell}|, \quad \text{SOPs}_\ell \triangleq \sum_{t=1}^T \sum_{i=1}^{N_\ell} |S_{t,\ell,i}| d_\ell \quad (51)$$

which reduces to the binary definitions when $m=1$. In MTE, integer codes appear only inside the refinement branch (Eq. (14)); the backbone itself remains a standard spike-driven network whose SR/SOPs are computed from its internal spike activity.

B.3. Accounting of non-spiking operations in MTE

Why we separate MTE MACs from spike proxies. MTE is a front-end module that operates on real-valued inputs before spiking computation. To avoid attributing its dense arithmetic to spike-based energy savings, we report (and include) MTE’s multiply–accumulate operations (MACs) separately when presenting overall accuracy–efficiency trade-offs.

Complexity of the refinement branch. For the 1D instantiation in Appendix B.1, the dominant dense cost comes from the per-channel temporal linear maps:

$$\text{MACs}(\text{Linear}_t^{(1)}) \approx B \cdot C \cdot L_t \cdot L_{t+1} \quad (52)$$

$$\text{MACs}(\text{Linear}_t^{(2)}) \approx B \cdot C \cdot L_{t+1} \cdot L_{t+1} \quad (53)$$

Because $L_{t+1} \leq L_t$ and L_t quickly decreases and is lower-bounded by $L_{\min}$, the added cost is typically modest under short horizons (e.g., $T=4$) compared with backbone computation. We empirically report the MAC breakdown (MTE vs. backbone) in Section 5.

Pooling/interpolation cost. Adaptive average pooling and interpolation mainly consist of additions and scalar multiplications by fixed coefficients. We count them in the MTE MAC budget for completeness, but they are minor relative to Eq. (53) and backbone MACs.

B.4. Why MTE is not a feature pyramid

Although MTE uses multi-scale resampling, it is not a feature pyramid (e.g., FPN) in either its role or its objective. The key differences are:

- **Different objective.** FPNs fuse multi-level *semantic features* to improve dense prediction (e.g., detection/segmentation), whereas MTE allocates a limited *time-step budget* by constructing a T -step *drive stream* for short-horizon spiking computation.
- **Different interface.** FPNs operate on intermediate backbone features across stages, while MTE is applied *before* the spiking backbone as an input-to-drive encoder, outputting $\{X^{(t)}\}_{t=1}^T$ as the synaptic drive in Eq. (1).
- **Different coupling mechanism.** FPNs couple scales via feature fusion across pyramid levels, whereas MTE couples scales via explicit per-step spike budgets $\{m_t\}$ and a short horizon T : each time step corresponds to a distinct temporal resolution, and multi-bit precision is allocated across steps.
- **Different outputs and evaluation.** FPNs output a set of feature maps for multi-scale heads, while MTE outputs a time-indexed drive stream and is evaluated by short-horizon accuracy–energy trade-offs.

In short, MTE is a *temporal coding* mechanism that uses scale to diversify evidence across a small number of time steps, rather than a semantic feature pyramid.

B.5. Additional discussion: why multi-scale helps under short horizons

Under a small simulation budget, naive repetition allocates multiple time steps to essentially the same observation. MTE instead introduces complementary drives at different temporal resolutions: higher-resolution steps preserve fine-grained local patterns, while lower-resolution steps capture longer-range context through aggregation. This complements Proposition 1 (Section 4.1): when time steps are scarce, improving per-step information efficiency is crucial for stabilizing short-horizon readout, and MTE provides a structured way to diversify evidence across $t = 1, \dots, T$.

C. Experimental Setup

C.1. Dataset details

We evaluate our framework on six temporal recognition tasks with **ten** public datasets: (i) modulation recognition (RML2016.10a, RML2016.10b) (O’Shea & West, 2016; O’Shea et al., 2016), (ii) ECG classification (MIT-BIH, PTB-XL) (Moody & Mark, 2001; Wagner et al., 2020), (iii) EEG emotion recognition (SEED) (Zheng & Lu, 2015; Duan et al., 2013), (iv) EMG gesture recognition (Ninapro.E1, Ninapro.E2), (v) mechanical fault diagnosis (CWRU4, CWRU10) (Smith & Randall, 2015), and (vi) speech/audio classification (SCD).

For each dataset, we follow the standard protocol when available. Unless otherwise stated, we use a 7:2:1 train/val/test split. For subject-dependent datasets (e.g., SEED and Ninapro), we additionally report subject-wise splits to reduce potential identity leakage. The exact preprocessing, segment length L , and split strategy for each dataset are summarized in Table 5.

Table 5. Dataset statistics after preprocessing. L denotes segment length.

Dataset	Task	Samples	Channels	L	Classes
RML2016a	Modulation	220,000	2	128	11
RML2016b	Modulation	1,200,000	2	128	11
MIT-BIH	ECG	83,598	1	187	5
PTB-XL	ECG	17,469	12	500	5
SEED	EEG	540	4	1,325	3
NinaproE1	EMG	6,418	10	200	12
NinaproE2	EMG	8,748	10	200	17
SCD	Speech	64721	64	32	30
CWRU4	Fault diagnosis	2,040	1	400	4
CWRU10	Fault diagnosis	8,400	1	400	10

C.2. Baseline details

We consider four backbone families for temporal modeling: (i) RNN/LSTM (Hochreiter & Schmidhuber, 1997), (ii) TCN (Bai et al., 2018), (iii) CNN backbones (LeNet/ResNet-style 1D CNNs) (LeCun et al., 1998; He et al., 2016), and (iv) Transformer-based models (Vaswani et al., 2017). For each backbone, we evaluate both an ANN and its SNN counterpart under the *matched-backbone*, *matched-budget* protocol (Section 3): identical macro-architecture and the same per-run training budget *without additional tricks*. The SNN counterpart is trained end-to-end with surrogate gradients using SpikingJelly (Fang et al., 2023), where the corresponding nonlinearities are replaced by multi-step spiking neurons and the network is unrolled for T steps. All backbone configurations are listed in Table 6.

C.3. Implementation details

Preprocessing. All models take **1D segments** as input. We apply per-channel normalization:

$$\tilde{\mathbf{x}} = \frac{\mathbf{x} - \mu}{\sigma}, \quad \mu = \mathbb{E}[\mathbf{x}], \quad \sigma = \sqrt{\text{Var}[\mathbf{x}]} \quad (54)$$

computed on the training set. (Optional) time–frequency representations are only used for an auxiliary ablation baseline and are not part of our main pipeline.

Training. We use Adam with batch size 64, initial learning rate 10^{-4} , and weight decay 5×10^{-4} . Early stopping is applied with patience 30 on the validation set. Unless otherwise stated, SNNs use a short horizon $T=4$. For STE we vary the replacement ratio $p \in \{0, 0.25, 0.5, 0.75, 1\}$ and ablate the transform width K ; for MTE we set $L_{\min}=32$ and use a decreasing multi-bit schedule (default $[4, 3, 2, 1]$ for $T=4$). We run five independent trials with different seeds and report the mean; standard deviations are omitted from the compact tables for space. Experiments are conducted on two NVIDIA RTX 4090 GPUs.

Metrics and efficiency accounting. We report ACC and Macro-F1. For efficiency, we report SR and SOPs (Section 3) and additionally account for inference-time dense MACs introduced by MTE and STE-analysis; these MACs are reported separately and included in total compute comparisons.

D. Additional Experiments

D.1. Extended Main Results

This section reports the *complete* main results corresponding to Table 1. For every backbone–dataset pair, we provide the full **ACC / F1 / Rating** triplet, where Rating is the same energy-related proxy defined in Section 3. Specifically, Tables 7–10 expand the main table by additionally including macro-F1, while keeping the same matched-backbone, matched-budget evaluation protocol. As a result, the **ACC** and **Rating** entries in these appendix tables are aligned with those reported in Table 1.

The extended results confirm all observations made in the main paper. First, the relative ordering between ANN, directly trained SNNs, and **Ours** (MTE×STE) remains consistent when evaluated with macro-F1, indicating that the reported accuracy gains are not driven by class imbalance or metric-specific artifacts. Second, improvements brought by MTE×STE persist across heterogeneous task domains (modulation, fault diagnosis, ECG, EMG, speech, and EEG) and across diverse backbone families, including recurrent, convolutional, and attention-based models.

We emphasize that the main paper focuses on ACC and Rating for clarity and space constraints, while the present

Table 6. Backbone configurations under the matched-backbone, matched-budget protocol. All ANN-SNN pairs share the same macro-architecture (depth/width/connectivity) and parameter count; the SNN version replaces nonlinearities with multi-step spiking neurons and is unrolled for T steps.

Family	Backbone	Input	Depth	Width	Params
RNN	RNN	1D	1	512	371724
RNN	LSTM (Hochreiter & Schmidhuber, 1997)	1D	1	512	1468428
TCN	TCN (Bai et al., 2018)	1D	4	C_{input}	1749964
CNN	1D LeNet-style (LeCun et al., 1998)	1D	5	C_{input}	21750
CNN	1D ResNet-style (He et al., 2016)	1D	18	C_{input}	3858892
Transformer	Transformer (ViT-1D style) (Vaswani et al., 2017)	1D	6	$\text{patch_size} = 32$	2556108

table serves as a comprehensive reference for reproducibility and detailed inspection. All results follow the same matched-backbone, matched-budget protocol described in Section 3 and Section C.

D.2. Additional Component Ablations

We report complete component ablations on additional backbones (including SLeNet) and full metrics (ACC/F1/R) in Table 11. The trends are consistent with the main paper: MTE improves temporal evidence at the cost of higher activity, STE regularizes firing, and combining them yields the most favorable accuracy-activity trade-off.

D.3. Full MTE horizon sweeps details

We report complete horizon T sweeps for MTE across datasets and backbones. Tables 12 and 13 summarize accuracy (ACC), F1-score (F1), and spike activity rate (R) for SLSTM and SLeNet, respectively. These results complement the representative trends shown in Figure 3 and confirm the non-monotonic dependence on T and task-dependent optimal horizons.

D.4. Additional Multi-bit Schedule Results

This section reports complete ablations of integer-spike allocation schedules for MTE. While the main paper focuses on SLSTM to illustrate the design trade-offs (Table 3), Table 14 provides full results for both SLSTM and SLeNet backbones. Overall, the trends observed in the main paper remain consistent across backbones: scale-aligned schedules yield robust accuracy-activity trade-offs under short horizons.

D.5. Full STE width sweeps (K)

D.5.1. STE DICTIONARY WIDTH K (SRNN)

Table 15 reports the full K sweep results on the SRNN backbone. K is reported as a multiple of the baseline channel dimension. We report classification accuracy (ACC), F1-score (F1), and average spike activity rate (rate).

D.5.2. STE DICTIONARY WIDTH K (SLeNet)

Table 16 reports the corresponding K sweep on the SLeNet backbone. While some datasets saturate in accuracy, spike activity still varies noticeably with K , highlighting the importance of controlling representational capacity even when accuracy appears stable.

E. Theoretical Power Consumption Estimation

This appendix describes how we estimate inference energy via synaptic operations (SOPs), which is the standard proxy for event-driven computation in SNNs. It complements Section 5.4, where we report compact energy summaries in the main text.

E.1. Synaptic Operation Counting (SOPs)

In SNNs, neural activations are binary spike events that gate synaptic computation. A commonly used efficiency proxy is the number of synaptic operations (SOPs), i.e., the number of triggered synaptic accumulations induced by spikes. Compared to ANN FLOPs, SOPs explicitly account for activity sparsity and thus better reflect event-driven inference on neuromorphic substrates.

Consider an SNN with L spiking layers and a discrete simulation horizon T . For layer $l \in \{1, \dots, L\}$ and timestep $t \in \{1, \dots, T\}$, let $N_{\text{spike}}^{(l,t)}$ denote the number of emitted spikes at that layer and timestep. Let $N_{\text{syn}}^{(l,t)}$ denote the number of effective synaptic connections (fan-in) that are activated by a spike event at layer l at timestep t . The layer-wise SOPs are defined as:

$$\text{SOPs}^{(l)} \triangleq \sum_{t=1}^T N_{\text{spike}}^{(l,t)} \cdot N_{\text{syn}}^{(l,t)} \quad (55)$$

The total SOPs of the network are:

$$\text{SOPs} \triangleq \sum_{l=1}^L \text{SOPs}^{(l)} = \sum_{l=1}^L \sum_{t=1}^T N_{\text{spike}}^{(l,t)} \cdot N_{\text{syn}}^{(l,t)} \quad (56)$$

Table 7. Extended main results for **RNN-family** (RNN and LSTM). We report ACC / F1 / Rating.

Backbone	Variant	RML2016a			RML2016b			CWRU4			CWRU10			MIT-BIH			PTB-XL			Nina-E1			Nina-E2			SCD			SEED		
		ACC	F1	R	ACC	F1	R	ACC	F1	R	ACC	F1	R	ACC	F1	R	ACC	F1	R	ACC	F1	R	ACC	F1	R	ACC	F1	R	ACC	F1	R
RNN	CNN	.374	.349	–	.458	.445	–	.731	.726	–	.693	.686	–	.995	.990	–	.678	.538	–	.735	.733	–	.715	.713	–	.725	.725	–	.896	.895	–
	SNN	.369	.346	.160	.424	.416	.248	.964	.964	.151	.904	.903	.158	.995	.990	.309	.677	.536	.269	.600	.598	.233	.643	.639	.274	.793	.794	.205	.881	.988	.320
	MTE	.431	.421	.254	.487	.474	.308	.944	.944	.271	.933	.933	.147	.994	.990	.348	.688	.548	.319	.703	.702	.266	.737	.736	.330	.832	.834	.230	.874	.875	.347
	Ours (MTE×STE)	.447	.436	.174	.487	.476	.227	.983	.983	.188	.921	.921	.082	.994	.989	.204	.685	.553	.082	.812	.811	.280	.815	.815	.407	.813	.814	.141	.891	.891	.306
LSTM	CNN	.395	.373	–	.479	.470	–	.947	.948	–	.896	.895	–	.995	.990	–	.682	.553	–	.720	.718	–	.734	.729	–	.777	.777	–	.904	.903	–
	SNN	.375	.415	.405	.488	.479	.439	.992	.992	.118	.851	.850	.334	.994	.989	.390	.685	.530	.497	.744	.742	.686	.678	.678	.695	.502	.496	.286	.881	.882	.619
	MTE	.431	.421	.254	.487	.474	.308	.944	.944	.271	.933	.933	.147	.994	.990	.348	.688	.548	.319	.703	.702	.266	.737	.736	.330	.832	.834	.230	.874	.875	.347
	Ours (MTE×STE)	.429	.410	.249	.482	.474	.304	.994	.994	.172	.931	.931	.206	.994	.989	.280	.670	.576	.249	.508	.508	.492	.571	.568	.552	.526	.526	.110	.859	.857	.580

 Table 8. Extended main results for **TCN**. We report ACC / F1 / Rating.

Variant	RML2016a			RML2016b			CWRU4			CWRU10			MIT-BIH			PTB-XL			Nina-E1			Nina-E2			SCD			SEED		
	ACC	F1	R	ACC	F1	R	ACC	F1	R	ACC	F1	R	ACC	F1	R	ACC	F1	R	ACC	F1	R	ACC	F1	R	ACC	F1	R	ACC	F1	R
CNN	.602	.605	–	.637	.635	–	.914	.914	–	.846	.846	–	.956	.926	–	.521	.280	–	.667	.666	–	.630	.629	–	.908	.899	–	.754	.757	–
SNN	.569	.577	.168	.596	.595	.205	.878	.878	.184	.846	.844	.167	.950	.950	.223	.515	.282	.171	.530	.528	.145	.549	.545	.144	.885	.886	.151	.674	.679	.189
MTE (4321)	.601	.586	.162	.634	.639	.194	.864	.864	.190	.860	.858	.178	.951	.918	.226	.519	.292	.147	.611	.610	.138	.589	.587	.140	.895	.897	.152	.756	.757	.181
Ours (MTE×STE)	.608	.611	.112	.647	.636	.087	.964	.964	.087	.904	.904	.103	.956	.927	.088	.554	.347	.085	.591	.590	.095	.592	.588	.111	.912	.913	.077	.800	.798	.105

Remarks on $N_{\text{syn}}^{(l,t)}$. For feedforward layers, $N_{\text{syn}}^{(l,t)}$ corresponds to the fan-in of the linear or convolutional operator (optionally multiplied by the number of output positions for convolution). For gated or multi-branch modules (e.g., spiking LSTMs or Transformer blocks), $N_{\text{syn}}^{(l,t)}$ aggregates the synaptic connections across gates/branches that participate in the forward computation at timestep t . In practice, we implement $N_{\text{syn}}^{(l,t)}$ as a deterministic quantity given the module structure, and thus the variability across inputs primarily comes from $N_{\text{spike}}^{(l,t)}$.

E.2. From SOPs to Energy

Given SOPs, we estimate inference energy by assuming a constant energy cost per synaptic operation on the target hardware:

$$E(\text{SNN}) \approx E_{\text{syn}} \cdot \text{SOPs} \quad (57)$$

where E_{syn} denotes the unit energy per triggered synaptic operation. For reference, prior hardware studies often report energy constants at the picojoule (pJ) scale. For example, at a 45 nm process node, a representative cost is $E_{\text{MAC}} \approx 4.6 \text{ pJ}$ per multiply-accumulate (MAC) and $E_{\text{AC}} \approx 0.9 \text{ pJ}$ per accumulate (AC). Because spike events are binary gates, synaptic computation on neuromorphic substrates is commonly modeled using AC-like costs; accordingly, we report energy using such constants when forming theoretical estimates.

For ANNs, a corresponding coarse estimate is

$$E(\text{ANN}) \approx E_{\text{MAC}} \cdot \text{FLOPs} \quad (58)$$

We emphasize that these estimates are intended as *relative* efficiency proxies for algorithmic comparison, rather than exact chip-level power measurements, as they abstract away implementation details such as memory access, routing, and hardware scheduling.

E.3. Implementation Notes

We measure $N_{\text{spike}}^{(l,t)}$ by counting emitted spikes at each spiking layer during inference, averaged over the evaluation set. We compute $N_{\text{syn}}^{(l,t)}$ from the module structure as the number of effective synaptic connections participating in the computation at timestep t . For multi-branch/gated modules, we sum the corresponding synaptic connections across gates/branches. When applicable, we report SOPs for the encoder and the backbone separately, to distinguish front-end encoding costs from backbone inference costs.

Notes on hardware realizations. On general-purpose GPUs, spike tensors are typically represented using floating-point formats, and the realized wall-clock speed/energy may deviate from the theoretical SOPs scaling. In contrast, neuromorphic processors can process spike events as Boolean streams and exploit event-driven scheduling. Therefore, SOPs-based estimates are most informative when comparing methods under a consistent implementation setting, and when interpreting potential efficiency benefits on event-driven hardware.

F. Limitations and Future Directions

This appendix discusses limitations of the current study and practical directions to strengthen the proposed MTE×STE framework and its evaluation.

F.1. Limitations

Scope of tasks and baselines. This study focuses on continuous sensory 1D temporal signal classification and does not claim general applicability to discrete sequence domains such as NLP or genomics, where dense embeddings and token-level semantics dominate the input representation. Our main comparisons follow a

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Table 9. Extended main results for **CNN-family** backbones (LeNet5-1D and ResNet18-1D). We report ACC / F1 / Rating.

Backbone	Variant	RML2016a			RML2016b			CWRU4			CWRU10			MIT-BIH			PTB-XL			Nina-E1			Nina-E2			SCD			SEED		
		ACC	F1	R	ACC	F1	R	ACC	F1	R	ACC	F1	R	ACC	F1	R	ACC	F1	R	ACC	F1	R	ACC	F1	R	ACC	F1	R	ACC	F1	R
LeNet5-1D	CNN	.492	.502	—	.521	.501	—	.979	.977	—	.996	.996	—	.990	.990	—	.660	.505	—	.625	.625	—	.645	.641	—	.811	.810	—	.770	.769	—
	SNN	.428	.425	.272	.472	.466	.343	.959	.951	.239	.992	.990	.296	.993	.987	.290	.649	.480	.218	.496	.498	.302	.545	.539	.329	.712	.712	.321	.704	.701	.216
	MTE (4321)	.446	.437	.291	.497	.471	.319	.984	.981	.232	.993	.991	.285	.995	.991	.260	.651	.482	.234	.505	.503	.242	.587	.588	.293	.707	.706	.311	.763	.762	.190
	Ours (MTE×STE)	.465	.467	.152	.496	.480	.156	.989	.986	.165	.997	.996	.166	.996	.993	.142	.653	.502	.125	.504	.499	.156	.588	.583	.182	.717	.716	.159	.770	.771	.124
ResNet18-1D	CNN	.576	.589	—	.639	.639	—	.996	.995	—	.966	.966	—	.994	.988	—	.705	.588	—	.900	.896	—	.896	.895	—	.932	.929	—	.815	.815	—
	SNN	.582	.597	.255	.623	.621	.250	.975	.975	.227	.960	.960	.238	.995	.992	.232	.694	.556	.219	.882	.882	.233	.847	.846	.236	.925	.925	.252	.778	.777	.235
	MTE (4321)	.581	.594	.215	.628	.626	.242	.999	.999	.237	.963	.962	.214	.996	.993	.224	.701	.540	.210	.893	.891	.206	.881	.880	.221	.925	.927	.244	.889	.889	.229
	Ours (MTE×STE)	.564	.575	.178	.626	.625	.165	.999	.999	.240	.968	.967	.204	.997	.995	.138	.706	.548	.156	.903	.901	.188	.900	.899	.206	.934	.934	.161	.919	.919	.207

Table 10. Extended main results for **Transformer-family** backbone (Spikformer-1D). We report ACC / F1 / Rating.

Variant	RML2016a			RML2016b			CWRU4			CWRU10			MIT-BIH			PTB-XL			Nina-E1			Nina-E2			SCD			SEED		
	ACC	F1	R	ACC	F1	R	ACC	F1	R	ACC	F1	R	ACC	F1	R	ACC	F1	R	ACC	F1	R	ACC	F1	R	ACC	F1	R	ACC	F1	R
CNN	.550	.553	—	.615	.611	—	.997	.997	—	.970	.970	—	.995	.991	—	.672	.503	—	.862	.861	—	.870	.870	—	.840	.842	—	.859	.857	—
SNN	.486	.481	.129	.560	.558	.172	.997	.997	.088	.976	.976	.141	.995	.991	.125	.672	.534	.062	.870	.870	.140	.867	.866	.165	.817	.815	.260	.815	.814	.123
MTE (4321)	.561	.560	.099	.569	.562	.169	.999	.999	.083	.972	.972	.107	.994	.989	.151	.678	.526	.056	.869	.868	.123	.865	.863	.163	.807	.807	.212	.807	.807	.108
Ours (MTE×STE)	.569	.557	.121	.561	.559	.093	.999	.999	.053	.983	.983	.094	.997	.995	.100	.676	.523	.039	.809	.808	.091	.833	.834	.101	.825	.824	.111	.844	.844	.085

Table 11. Component ablation on two backbones.

Backbone Variant		SEED		MIT-BIH		CWRU10		AVG	
		ACC	R	ACC	R	ACC	R	ACC	R
SLSTM	ANN	.904	—	.995	—	.896	—	.932	—
	SNN	.896	.553	.994	.213	.832	.152	.907	.306
	STE	.951	.439	.995	.207	.851	.133	.932	.260
	MTE	.904	.619	.994	.290	.911	.234	.936	.381
	MTE+STE	.899	.530	.994	.180	.931	.206	.941	.305
SLeNet	ANN	.770	—	.995	—	.996	—	.920	—
	SNN	.704	.216	.993	.290	.992	.296	.896	.267
	STE	.746	.109	.993	.123	.990	.157	.910	.130
	MTE	.763	.190	.995	.260	.993	.285	.917	.245
	MTE+STE	.770	.124	.996	.142	.997	.166	.921	.144

Table 12. **MTE horizon T sweep on SLSTM.** ACC / F1 / activity rate (R).

T	SEED			CWRU10			Nina-E1			PTB-XL		
	ACC	F1	R	ACC	F1	R	ACC	F1	R	ACC	F1	R
1	.858	.857	.356	.845	.846	.183	.505	.507	.312	.655	.515	.355
4	.872	.871	.460	.892	.892	.226	.545	.543	.617	.662	.528	.341
8	.874	.874	.469	.915	.915	.262	.572	.571	.658	.668	.522	.455
12	.868	.868	.466	.904	.903	.300	.566	.562	.593	.670	.524	.427
16	.852	.852	.554	.905	.905	.148	.548	.546	.500	.668	.522	.469

Table 13. **MTE horizon T sweep on SLeNet.** ACC / F1 / activity rate (R).

T	SEED			CWRU10			Nina-E1			PTB-XL		
	ACC	F1	R	ACC	F1	R	ACC	F1	R	ACC	F1	R
1	.688	.683	.028	.945	.944	.037	.395	.390	.338	.640	.500	.030
4	.695	.690	.033	.973	.973	.029	.520	.517	.254	.664	.530	.027
8	.702	.698	.072	.975	.974	.057	.540	.537	.269	.666	.520	.034
12	.704	.704	.069	.981	.981	.050	.545	.541	.277	.668	.524	.052
16	.698	.694	.070	.979	.979	.038	.538	.534	.287	.664	.518	.033

Table 14. **MTE multi-bit schedule ablation (full).** ACC and activity rate (R) under different integer-spike allocation schedules.

Backbone	Dataset	1111		1234		4321		4444	
		ACC	R	ACC	R	ACC	R	ACC	R
SLSTM	CWRU4	.994	.053	.992	.061	.992	.058	.994	.054
	SEED	.867	.656	.852	.634	.859	.674	.867	.585
	PTB-XL	.659	.375	.670	.386	.674	.393	.658	.361
	Nina-E1	.588	.600	.569	.614	.589	.632	.581	.645
	Nina-E2	.589	.648	.589	.671	.578	.629	.585	.623
SLeNet	CWRU4	.994	.123	.994	.124	.997	.121	.992	.121
	SEED	.837	.332	.852	.348	.867	.324	.852	.349
	PTB-XL	.674	.307	.680	.298	.680	.311	.678	.303
	Nina-E1	.619	.253	.627	.263	.626	.276	.638	.279
	Nina-E2	.644	.298	.663	.295	.645	.297	.651	.302

strict matched-backbone/matched-budget protocol to isolate spiking-specific effects. While this offers controlled and reproducible evidence for the ANN-SNN gap phenomenon and the benefits of MTE×STE, it does not exhaustively cover task-specific pipelines, highly customized architectures, or compressed ANN variants such as BNNs and INT4/INT8 temporal models. As a result, the reported numbers should be interpreted as a principled, controlled assessment rather than a comprehensive leaderboard study.

Efficiency evaluation. Our efficiency analysis follows standard SOP/SR-based algorithmic proxies, which are widely used for estimating spike-driven computational savings. However, these proxies do not include chip-level memory access, NoC routing, scheduling, cross-core communication, or hardware-specific data-movement overheads. Therefore, the reported efficiency trends mainly reflect algorithmic sparsity and spike operations, while full hardware validation on neuromorphic or event-driven accelerators remains an important future direction.

Partial mechanistic coverage. Section 4.1 provides a

Table 15. STE dictionary width K sweep on SRNN. Moderate K often improves accuracy while maintaining lower spike activity.

K	CWRU4			SEED			PTB-XL			Ninapro-E1			Ninapro-E2		
	ACC	F1	rate	ACC	F1	rate	ACC	F1	rate	ACC	F1	rate	ACC	F1	rate
1×	.883	.883	.145	.919	.919	.040	.682	.547	.022	.747	.744	.023	.773	.771	.024
2×	.975	.975	.040	.933	.932	.082	.693	.552	.015	.759	.757	.028	.742	.739	.026
3×	.669	.670	.284	.904	.904	.041	.685	.553	.013	.778	.775	.016	.753	.752	.067
4×	.831	.829	.302	.933	.933	.125	.691	.559	.014	.764	.762	.028	.754	.751	.030

 Table 16. STE dictionary width K sweep on SLeNet. Even when accuracy saturates, spike activity remains sensitive to K .

K	CWRU4			SEED			PTB-XL			Ninapro-E1			Ninapro-E2		
	ACC	F1	rate	ACC	F1	rate	ACC	F1	rate	ACC	F1	rate	ACC	F1	rate
1×	1.000	1.000	.084	.696	.692	.031	.647	.491	.031	.548	.545	.087	.511	.511	.080
2×	1.000	1.000	.094	.704	.706	.063	.662	.520	.094	.718	.716	.094	.722	.721	.065
3×	1.000	1.000	.093	.763	.761	.075	.668	.508	.075	.744	.743	.051	.734	.733	.116
4×	1.000	1.000	.072	.815	.813	.100	.669	.522	.053	.748	.750	.053	.750	.749	.079

mechanism-level explanation centered on short-horizon evidence accumulation and finite-window encoding uncertainty. However, real continuous 1D temporal signals can exhibit additional structures, such as non-stationarity, heavy-tailed bursts, or domain-specific preprocessing, that are not explicitly modeled in our analysis. Therefore, while our proposition motivates design choices for the short-horizon regime, it does not fully characterize all sources of performance variation across tasks and datasets.

Sensitivity to design choices. Although MTE and STE are plug-and-play, their behavior depends on several design knobs, including the horizon T , multi-bit schedule, STE width K , and replacement ratio p . We provide ablations and recommend default settings, but a universally optimal configuration is unlikely across heterogeneous domains. In the current study, the STE replacement ratio is fixed after a separate backbone-level replacement study to avoid an additional model-selection advantage in the main comparisons. Nevertheless, adaptive layer-wise or input-dependent replacement policies may be necessary for deeper residual or Transformer-style backbones. Moreover, the training-only reconstruction branch in STE introduces additional training-time overhead, which may be non-negligible in large-scale settings even though it does not affect inference.

F.2. Future Directions

Broader and standardized evaluation. A useful next step is to expand evaluation to larger-scale and more diverse 1D corpora under standardized splits and reporting, and to release unified training recipes for matched-budget comparisons. This would further clarify when the small ANN-SNN gap holds and how it depends on horizon length and data regimes.

Adaptive resource allocation. MTE currently uses a fixed scale schedule and STE uses a fixed replacement ratio.

An important direction is to make these allocations data-dependent, e.g., learning per-sample or per-layer schedules for temporal scales and sparsity budgets, potentially improving robustness under non-stationary or mixed-dependency signals.

Bridging proxies to measured deployment metrics. We quantify efficiency using spike-based proxies (SR/SOPs) under explicit accounting assumptions. Future work should validate these trends on representative neuromorphic or embedded platforms, and study how architectural choices (e.g., multi-bit spikes and transform coding) translate to latency/energy under realistic system constraints.